

Slip 16

Q.1) Display all the files from current directory which are created in particular month
[10 Marks]

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <sys/stat.h>
#include <dirent.h>
#include <time.h>

int main()
{
    DIR *dp;
    struct stat s;

    struct dirent *entry;
    char month[10]; // The month user will input
    char *date;     // To store date string
    printf("Enter the month: ");
    scanf("%s", month); // Corrected the scanf here, no need for '&'

    dp = opendir("."); // Open the current directory
    if (dp == NULL)
    {
        printf("Sorry, directory not open\n");
        return 1;
    }
    else
    {
        printf("Files created in the month '%s':\n", month);
        // Loop through all entries in the directory
        while ((entry = readdir(dp)) != NULL)
        {
            if (stat(entry->d_name, &s) == 0)
            {
                date = ctime(&s.st_ctime); // Get the creation date of the
file

                // The ctime returns date format like: "Thu Oct 21 14:25:30
2021"

                // Extracting the first three letters of the month part (e.g.,
"Oct" from "Thu Oct 21...")
                if (strstr(date, month))
                {
                    // Print the file name and the full date
                    printf("File: %s, Date: %s", entry->d_name, date);
                }
            }
        }
    }
}
```

```

        }
    }
    closedir(dp); // Close the directory after we're done
}

return 0;
}

}
struct tm *tm_info = localtime(&st.st_mtime);
if ((tm_info->tm_mon + 1) == month) { // tm_mon: 0-11
printf("%s\n", entry->d_name);
}
}
closedir(dir);
return 0;
}

```

Q.2) Write a C program which create a child process which catch a signal sighup, sigint and sigquit.

The Parent process send a sighup or sigint signal after every 3 seconds, at the end of 30 second

parent send sigquit signal to child and child terminates my displaying message "My DADDY has

Killed me!!!". [20 Marks]

```

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <signal.h>
#include <sys/wait.h>
void handle_signal(int sig) {
if (sig == SIGHUP) {
printf("Child received SIGHUP\n");
} else if (sig == SIGINT) {
printf("Child received SIGINT\n");
} else if (sig == SIGQUIT) {
printf("My DADDY has Killed me!!!\n");
exit(0);
}
}
int main() {
pid_t pid = fork();
if (pid < 0) {
perror("fork");
exit(1);
}
if (pid == 0) {
// Child process: setup signal handlers
signal(SIGHUP, handle_signal);

```

```
signal(SIGINT, handle_signal);
signal(SIGQUIT, handle_signal);
while (1) {
    pause(); // Wait for signals
}
} else {
    // Parent process
    int count = 0;
    while (count < 10) { // 10 * 3 sec = 30 sec
        sleep(3);
        if (count % 2 == 0) {
            kill(pid, SIGHUP);
        } else {
            kill(pid, SIGINT);
        }
        count++;
    }
    // After 30 seconds, terminate child
    kill(pid, SIGQUIT);
    wait(NULL);
    printf("Parent process exiting.\n");
}
return 0;
}
```